7.7 SPI Secondary Device Mode Timings

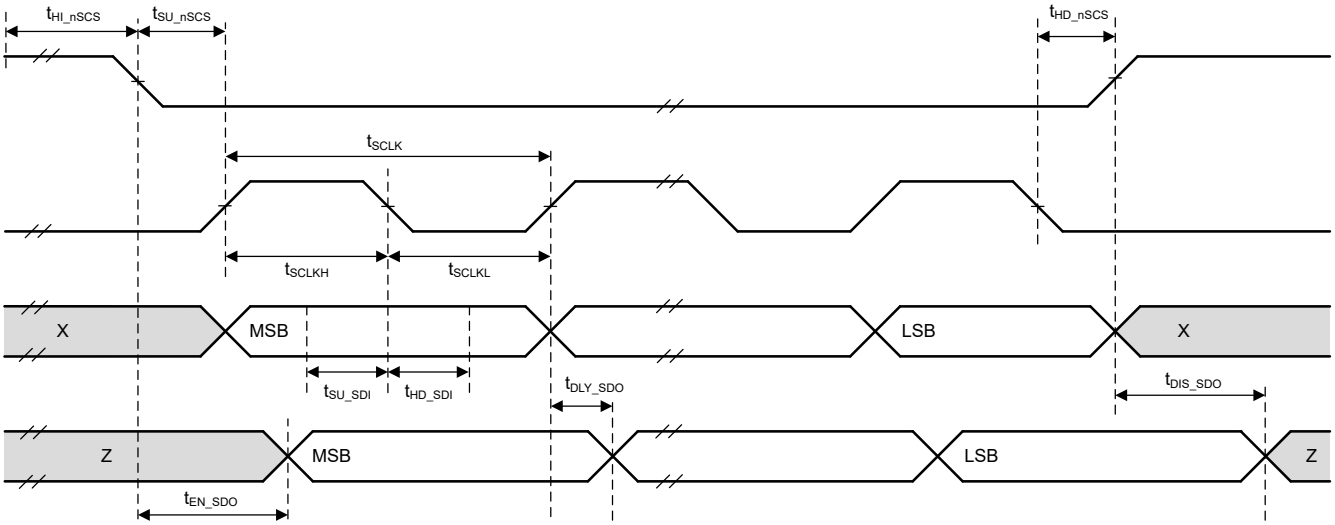


Figure 7-1. SPI Secondary Device Mode Timing Diagram

8.3.7 Cross Conduction (Dead Time)

The device is fully protected from cross conduction of MOSFETs. The high-side and low-side MOSFETs are operated to avoid any shoot through currents by inserting a dead time (t_{DEAD}). This is implemented by sensing the gate-source voltage (VGS) of the high-side and low-side MOSFETs and ensured that VGS of high-side MOSFET has reached below turn-off levels before switching on the low-side MOSFET of same half-bridge as shown in Figure 8-20 and Figure 8-21. The VGS of the high-side and low-side MOSFETs (VGS_HS and VGS_LS) shown in Figure 8-21 are DRV8311 internal signals.

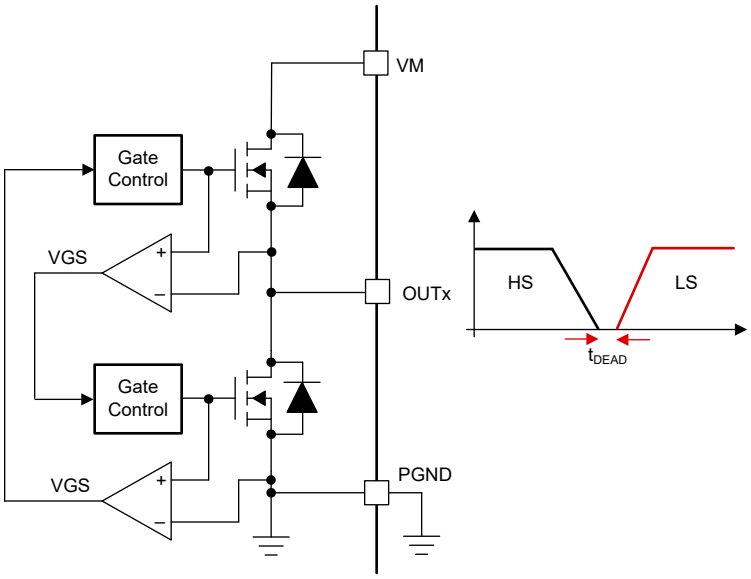


Figure 8-20. Cross Conduction Protection

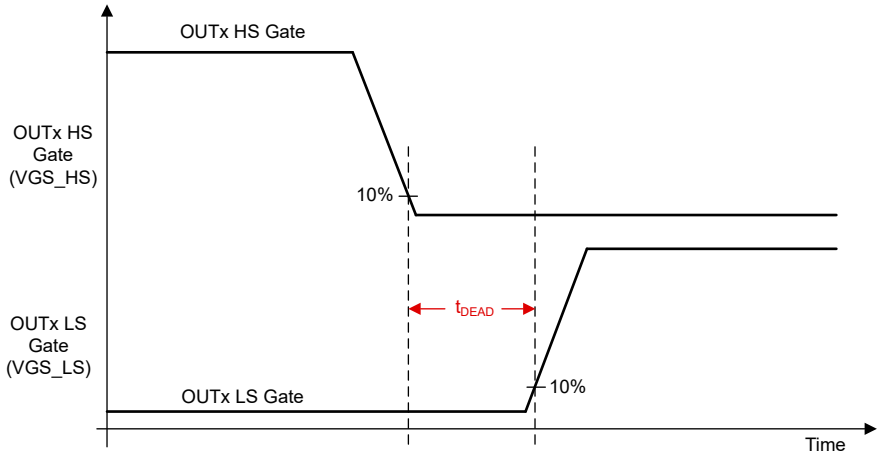


Figure 8-21. Dead Time

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